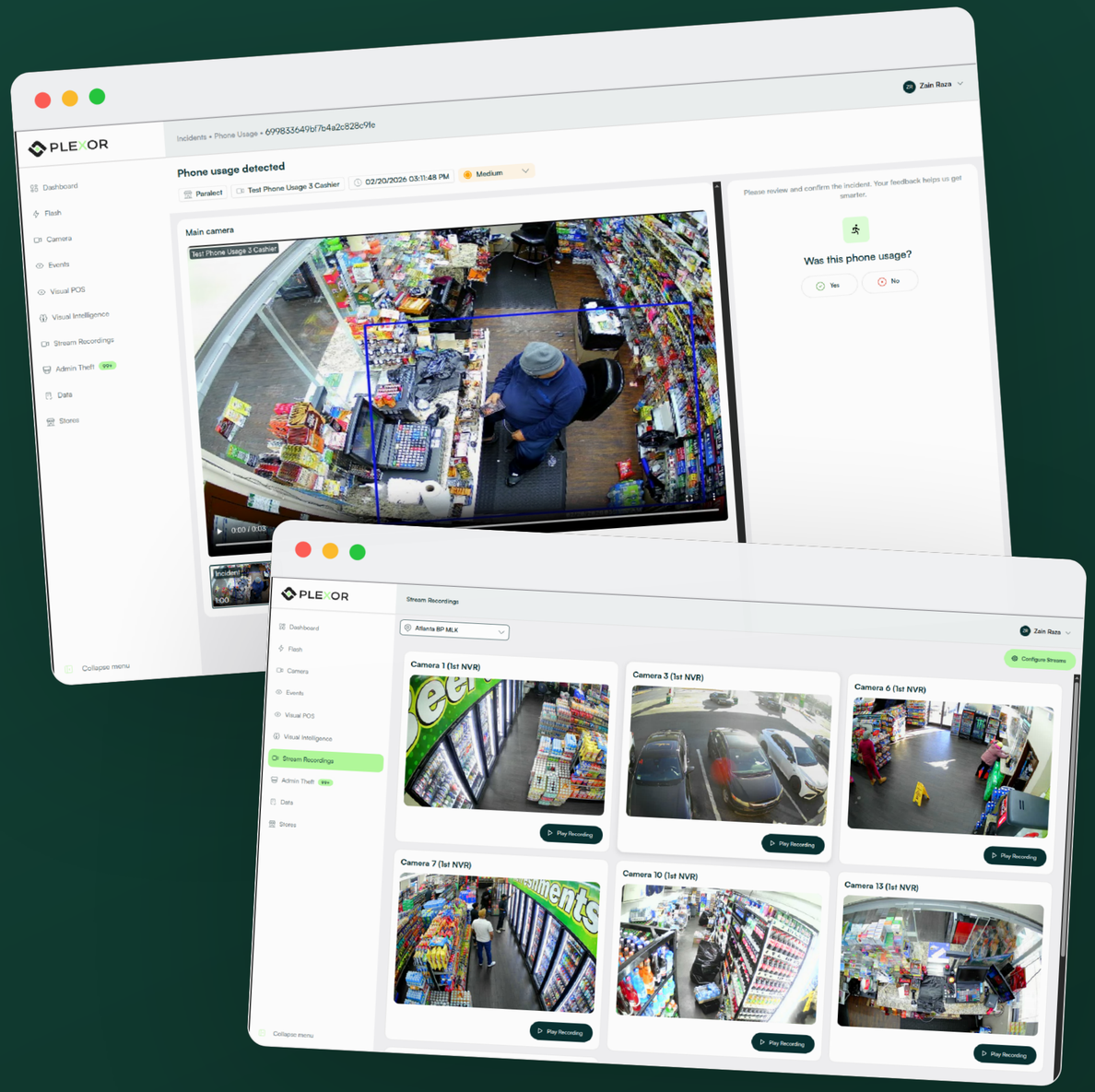
PLEXOR

01 / AI SURVEILLANCE

AI-Based Surveillance System

Computer-vision loss-prevention for retail — theft, POS fraud and loitering, flagged in real time.



Zohaib Ali Mughal — Full Stack Developer

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01 — OVERVIEW

AI-Based Surveillance System

Retail shrinkage from employee & customer theft and POS under-ringing is hard to catch live. A multi-camera computer-vision platform detects it and routes alerts to store and district managers.

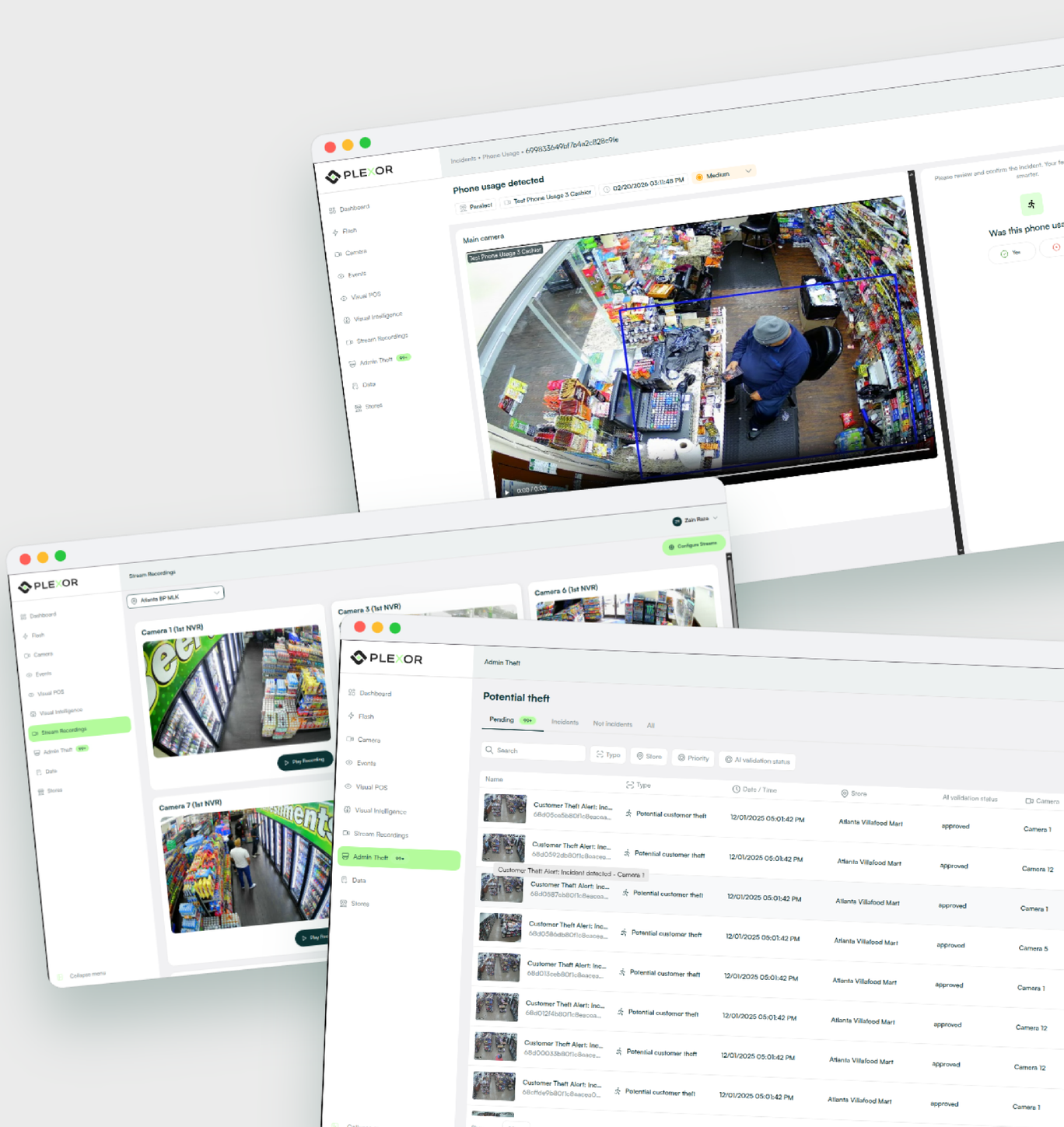
ROLE **Full Stack Developer**

100+
STORES MONITORED

~700ms
REALTIME INFERENCE

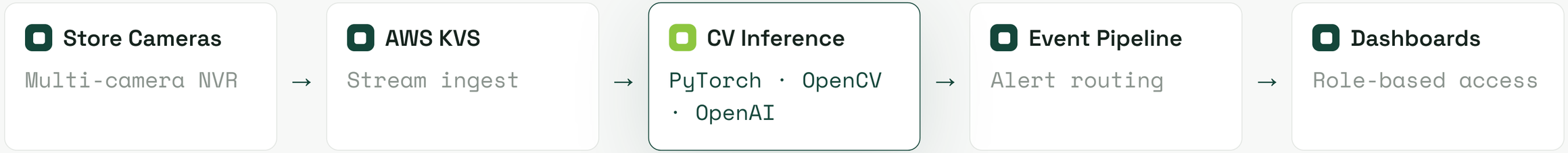
37+
DETECTION MODELS

- Theft detection across employees & customers
- Phone-usage & loitering alerts with live recording streams
- Visual POS & under-ringing detection



01 — ARCHITECTURE & STACK

Realtime vision pipeline, role-scoped delivery



STACK

FRONTEND	Next.js · Mantine
BACKEND	Node.js · Fastify · Python
CLOUD	AWS · EKS · Lambda · S3 · KVS
AI	PyTorch · OpenCV · OpenAI

KEY DECISIONS

01 Realtime multi-camera streaming over AWS KVS for low-latency feeds

02 Model serving on EKS for scalable GPU inference

03 Role-scoped dashboards for store, district & admin views

01 — WALKTHROUGH

Inside the product

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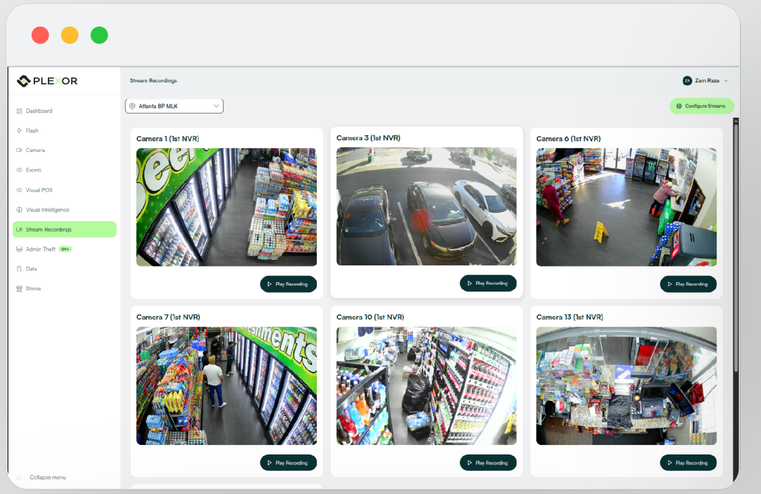
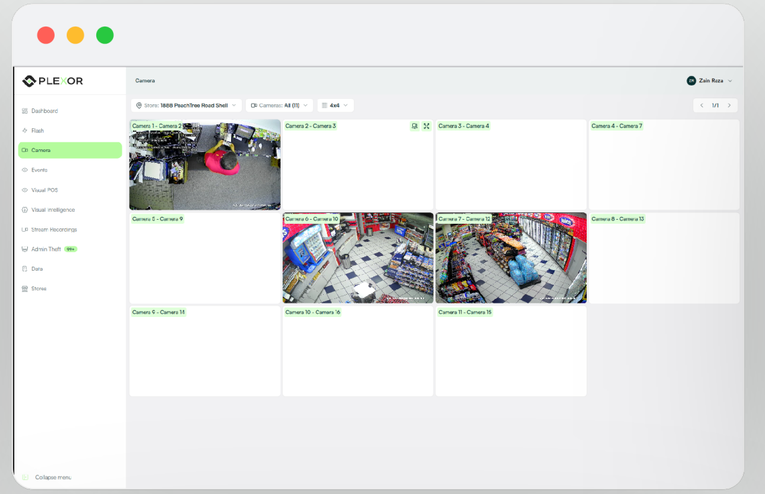
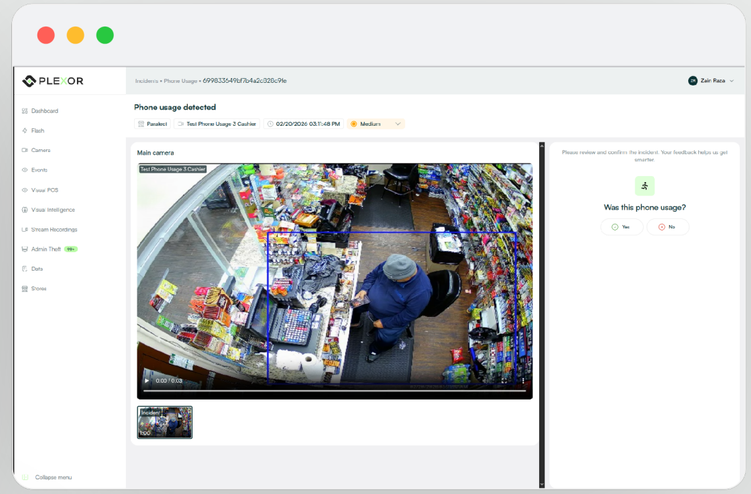


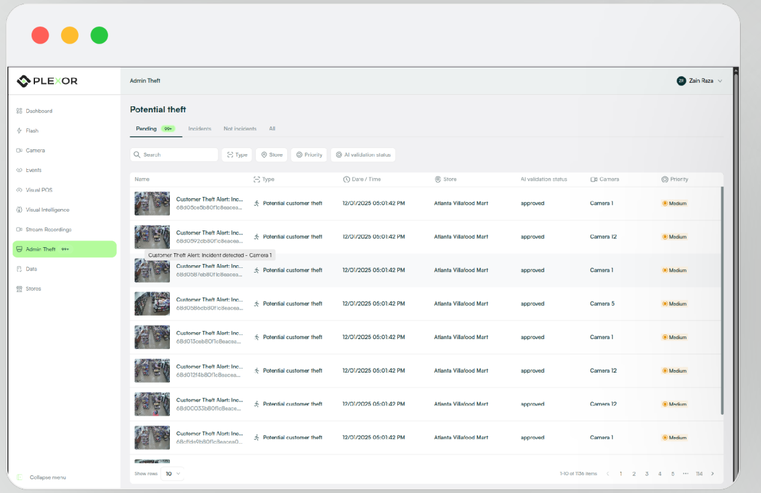
Live recording streams



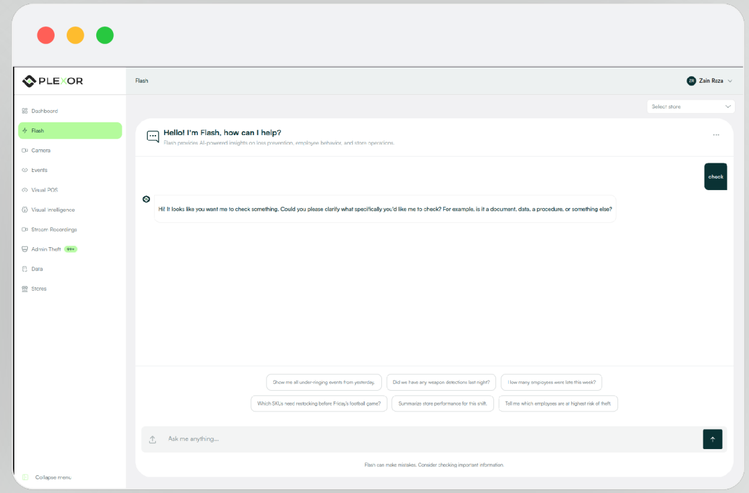
Multi-camera live grid



Phone-usage detection & review



Theft queue & AI validation



Flash — AI loss-prevention assistant

01 — IMPACT

Loss prevention, measured

100+
STORES MONITORED

~700ms
REALTIME INFERENCE

37+
DETECTION MODELS

Surfaced theft and POS under-ringing events in real time across monitored stores, routing prioritised alerts to the managers who could act on them.

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